# Frameworks

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### Open-Source Frameworks:

AA

| Framework | Focus | Key Features | Applications | Strengths | Limitations |
|---|---|---|---|---|---|
| **Retrieval-Augmented Generation Architectures (RAGAs)** | Evaluation by combining LLMs with retrieval systems to enhance factuality and relevance | Retrieves data from external sources during response generation | Applications requiring current or domain-specific knowledge, e.g., customer support, legal, medical | Reduces hallucinations, improves accuracy by grounding in real-time data | Performance depends on retrieval quality and data source accuracy |
| **VADER** | Good solution for sentiment analysis of short text | • Rule-based and lexicon approach with valence scores for individual words<br>• Handles degree modifiers, negation, capitalization, and punctuation<br>• Outputs scores for positive, neutral, negative, and compound (overall) sentiment | Can apply to dimensions: toxicity detection, bias detection, tone identification | Performs well on social media posts, informal text | • Performance depends on retrieval quality and data source accuracy<br>• Not as useful on long texts |
| **TruLens** | Open-source tool for evaluation, monitoring, and observability | Tracks metrics like relevance, accuracy, fairness, and provides observability features | Continuous monitoring and debugging of production LLMs | Customizable, provides robust feedback loops for model improvement | Requires configuration; complex for large-scale systems |
| **PromptFoo** | Evaluation framework for prompt optimization and A/B testing | Allows for prompt experimentation, scoring, and benchmarking across different prompt variations | Fine-tuning and debugging of prompts for specific use cases | Facilitates iterative prompt tuning and side-by-side comparison | Limited to prompt evaluation, not full-model performance |
| **MLflow** | Manages model tracking, experimentation, and deployment for ML and LLMs | Tracks experiments, metrics, and artifacts across LLM projects | Large-scale LLM lifecycle management in production | Excellent for reproducibility and lifecycle tracking | General ML tool, not specialized for LLMs |
| **DeepEval** | Framework for multi-metric evaluation, including qualitative metrics like interpretability | Supports detailed qualitative analysis of LLMs, including bias, fairness, and interpretability | Applications where interpretability and ethical concerns are high | Comprehensive qualitative metrics for in-depth LLM evaluation | Early-stage, may lack support for complex production workflows |
| **OpenAI Evals** | Open-source framework for custom LLM benchmarking | Allows design of custom benchmarks and use-case-specific evaluations | Customizable for specific LLM use cases like support chatbots, QA systems | Extensible, good for specific case-based benchmarking | Still evolving; limited pre-built benchmarks |

LCB

### Proprietary Frameworks:

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|---|---|---|---|---|---|
| **ChainPoll** | Evaluation of LLMs through stepwise reasoning and consistency polling | Combines chain-of-thought prompting with multiple response polling; averages results to gauge confidence and uncertainty | Evaluating LLM output quality, especially in contexts requiring rationale and consistency checks | Captures nuanced model behavior, provides confidence scores, scales well across contexts | **Proprietary**, high compute resources for multiple response polling, less effective for simple tasks |
| **Amazon Bedrock** | Managed service to evaluate, deploy, and customize multiple foundation models | Access to models from Amazon Titan, Anthropic, AI21 Labs; supports fine-tuning and prompt customization | Multi-model applications in e-commerce, finance, real-time insights | Simplifies customization, deployment, and scaling within AWS | Limited to AWS ecosystem; may lack niche models for specific tasks |
| **NVIDIA NeMo** | Evaluation framework for training and fine-tuning LLMs on custom datasets | Offers model training, fine-tuning, and evaluation with synthetic data generation | Custom language tasks, especially in voice and text applications | High compatibility with NVIDIA hardware, supports domain-specific data fine-tuning | Requires NVIDIA infrastructure and expertise in model fine-tuning |
| **Vertex AI Studio** | Google's evaluation and fine-tuning platform for LLMs | Offers a model comparison dashboard, tuning capabilities, and built-in metrics | Large-scale LLM deployments with integration into Google Cloud | Simplifies evaluation, tuning, and deployment with Google Cloud resources | Limited to Google Cloud infrastructure |
| **Azure AI Studio** | Microsoft's platform for evaluating and deploying LLMs within Azure ecosystem | Offers evaluation tools, model diagnostics, and deployment options | Enterprise-level LLM projects, often within Microsoft ecosystems | Strong integration with Microsoft services, provides deployment tools | Limited to Azure infrastructure, may not support non-Microsoft models |
| **IBM Foundation Model Evaluation (FM-eval)** | IBM's framework for evaluation on enterprise metrics like fairness, safety, and interpretability | Evaluates model fairness, interpretability, robustness, and domain-specific knowledge | Enterprises needing compliance-focused or highly interpretable models | Focused on ethical and enterprise metrics, useful for sensitive applications | Primarily suited for IBM environments or IBM-specific models |
| **Arize Phoenix** | Primarily evaluates the accuracy and relevance of information retrieval in LLM-generated answers to ensure factual reliability | Customizable templates for assessing response relevance, hallucination detection, and QA correctness in retrieval-heavy tasks | Best suited for applications like customer support, educational tools, and research assistants that depend on accurate information retrieval | Excels in tracking retrieval accuracy, handling complex query responses, and reducing hallucinations, which makes it ideal for QA and RAG applications | Limited flexibility beyond RAG-specific metrics and may require additional (you must pay for) tools for broader LLM evaluation |